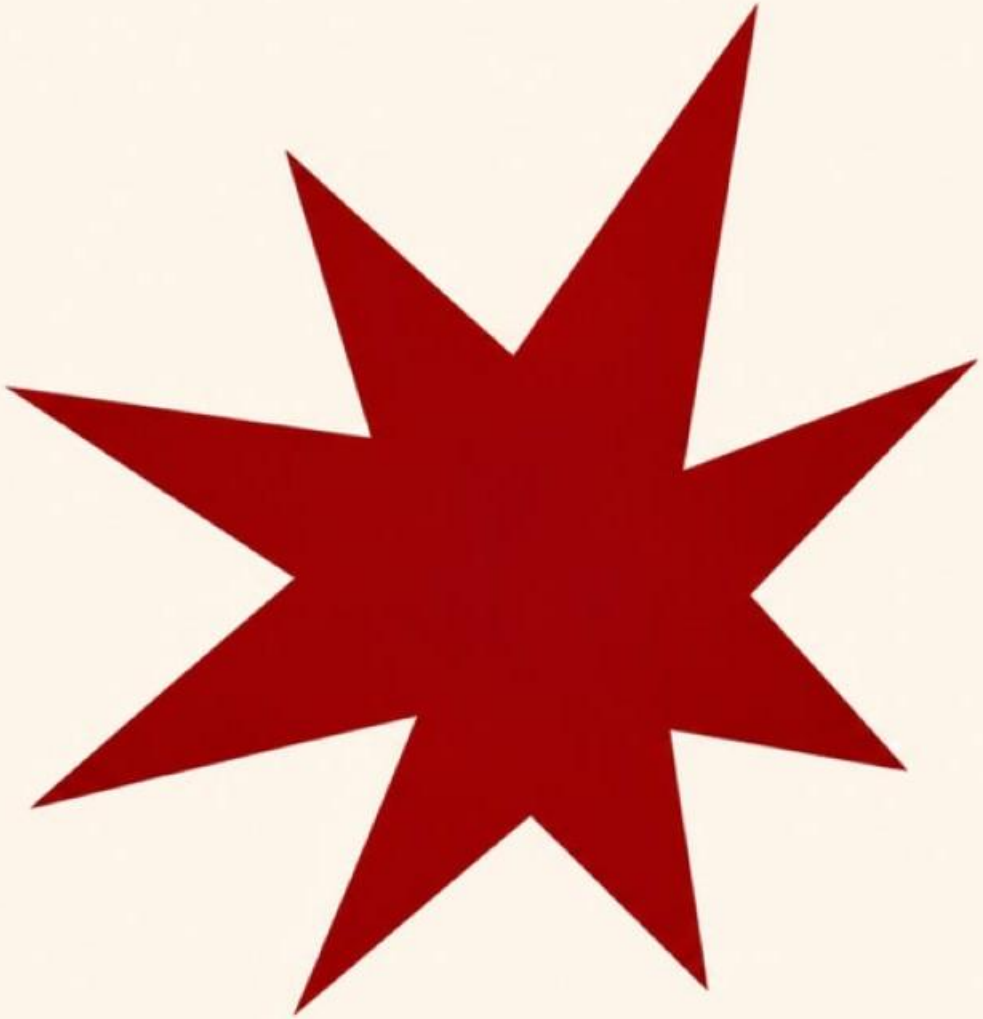
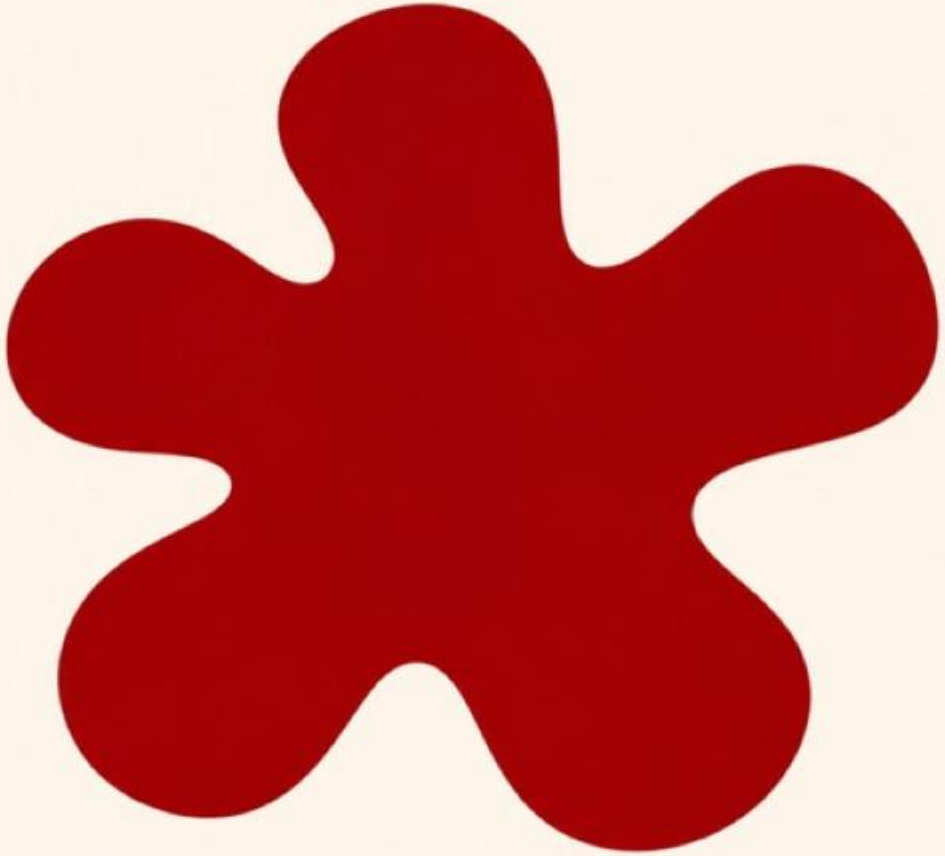
Trust Me, Bro

SOCIAL ENGINEERING
THE AI-AGENT LOOP



Danita Samuel Prakash





RED BLUE GREEN



Cognitive Theory



Authority Bias



Fear and Urgency



Social Engineering

The AI-Agent Loop

How social engineering can exploit both the AI and the human in the loop



Attacker

Manipulates information (e.g. a malicious email or document)



Untrusted Input

Malicious or persuasive instructions are introduced



AI Agent

Processes the input, interprets it and generates a recommendation



Human Approval

Reviews the recommendation and authorises the action



Potential Compromise

The attacker's objective is achieved (e.g. data access, unauthorised action or system change)

The Loop

The attacker exploits **trust, context** and **human behaviour** across the entire **AI-agent** workflow.